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Date	18-Oct-2025

GitHub repo - [SoCD Programming lab](#)

CS5107 Programming Lab

Assignment 2

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$ stat -c "Modified: %y File: %n" *
```

Modified: 2025-10-17 23:20:32.000000000 +0530	File: add.c
Modified: 2025-10-18 15:22:33.172452233 +0530	File: DeepMaths.c
Modified: 2025-10-17 23:26:40.000000000 +0530	File: div.c
Modified: 2025-10-18 15:24:36.196068272 +0530	File: headers.h
Modified: 2025-10-18 15:21:30.549393628 +0530	File: logarithm.c
Modified: 2025-10-18 18:49:35.055514414 +0530	File: Makefile
Modified: 2025-10-17 23:21:22.000000000 +0530	File: mul.c
Modified: 2025-10-18 15:24:23.530424514 +0530	File: power.c
Modified: 2025-10-18 15:25:58.462291304 +0530	File: root.c
Modified: 2025-10-18 14:29:19.909045431 +0530	File: SimpleMath.c
Modified: 2025-10-17 23:21:26.000000000 +0530	File: sub.c
Modified: 2025-10-18 15:28:50.607994633 +0530	File: UserInterface.c

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$
```

Figure 1: Initial file modified times

This image gives the modified time of each files.

After making small modification to the print statement inside SimpleMath.c file, the modified time become as,

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$ stat -c "Modified: %y File: %n" *
```

Modified: 2025-10-17 23:20:32.000000000 +0530	File: add.c
Modified: 2025-10-18 15:22:33.172452233 +0530	File: DeepMaths.c
Modified: 2025-10-17 23:26:40.000000000 +0530	File: div.c
Modified: 2025-10-18 15:24:36.196068272 +0530	File: headers.h
Modified: 2025-10-18 15:21:30.549393628 +0530	File: logarithm.c
Modified: 2025-10-18 18:49:35.055514414 +0530	File: Makefile
Modified: 2025-10-17 23:21:22.000000000 +0530	File: mul.c
Modified: 2025-10-18 15:24:23.530424514 +0530	File: power.c
Modified: 2025-10-18 15:25:58.462291304 +0530	File: root.c
Modified: 2025-10-18 19:17:59.428685528 +0530	File: SimpleMath.c
Modified: 2025-10-17 23:21:26.000000000 +0530	File: sub.c
Modified: 2025-10-18 15:28:50.607994633 +0530	File: UserInterface.c

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$
```

Figure 2: File modified time after changing 1 file

Doing “make” adds new .o files for the corresponding .c files.

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$ make
gcc -c UserInterface.c
gcc -c SimpleMath.c
gcc -c DeepMaths.c
gcc -c add.c
gcc -c sub.c
gcc -c mul.c
gcc -c div.c
gcc -c power.c
cc -c -o root.o root.c
gcc -c logarithm.c
gcc UserInterface.o SimpleMath.o DeepMaths.o add.o sub.o mul.o div.o power.o root.o logarithm.o -lm -o UserInterface
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$ stat -c "Modified: %y File: %n" *
Modified: 2025-10-17 23:20:32.000000000 +0530 File: add.c
Modified: 2025-10-18 19:24:51.848326612 +0530 File: add.o
Modified: 2025-10-18 15:22:33.172452233 +0530 File: DeepMaths.c
Modified: 2025-10-18 19:24:51.825326615 +0530 File: DeepMaths.o
Modified: 2025-10-17 23:26:40.000000000 +0530 File: div.c
Modified: 2025-10-18 19:24:51.919326605 +0530 File: div.o
Modified: 2025-10-18 15:24:36.196068272 +0530 File: headers.h
Modified: 2025-10-18 15:21:30.549393628 +0530 File: logarithm.c
Modified: 2025-10-18 19:24:52.021326594 +0530 File: logarithm.o
Modified: 2025-10-18 18:49:35.055514414 +0530 File: Makefile
Modified: 2025-10-17 23:21:22.000000000 +0530 File: mul.c
Modified: 2025-10-18 19:24:51.895326607 +0530 File: mul.o
Modified: 2025-10-18 15:24:23.530424514 +0530 File: power.c
Modified: 2025-10-18 19:24:51.953326601 +0530 File: power.o
Modified: 2025-10-18 15:25:58.462291304 +0530 File: root.c
Modified: 2025-10-18 19:24:51.987326598 +0530 File: root.o
Modified: 2025-10-18 19:24:35.593412444 +0530 File: SimpleMath.c
Modified: 2025-10-18 19:24:51.789326619 +0530 File: SimpleMath.o
Modified: 2025-10-17 23:21:26.000000000 +0530 File: sub.c
Modified: 2025-10-18 19:24:51.871326610 +0530 File: sub.o
Modified: 2025-10-18 19:24:52.056723112 +0530 File: UserInterface
Modified: 2025-10-18 15:28:50.607994633 +0530 File: UserInterface.c
Modified: 2025-10-18 19:24:51.754326622 +0530 File: UserInterface.o
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$
```

Figure 3: New files added after make and their modified times shown

Code runs good.

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$ ./UserInterface
Enter your choice
1. Normal calculation
2. Deep maths
3. Exit
1
Normal calculation
Enter your choice
1. Add
2. Sub
3. Mul
4. Div
5. Go back
1
Enter 2 numbers (with space): 50 20
Inside Add function
Result of calculation: 70.000000
Enter your choice
1. Normal calculation
2. Deep maths
3. Exit
3
Exiting...
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$
```

Figure 4: Code Execution with the added debug line

After a small print line to the `logarithm.c` and running “make” again, the Makefile does the compilation just for the newly changed file.

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$ make
gcc -c logarithm.c
gcc UserInterface.o SimpleMath.o DeepMaths.o add.o sub.o mul.o div.o power.o root.o logarithm.o -lm -o UserInterface
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$ stat -c "Modified: %y File: %n" *
```

File	Modified
add.c	2025-10-17 23:20:32.000000000 +0530
add.o	2025-10-18 19:24:51.848326612 +0530
DeepMaths.c	2025-10-18 15:22:33.172452233 +0530
DeepMaths.o	2025-10-18 19:24:51.825326615 +0530
div.c	2025-10-17 23:26:40.000000000 +0530
div.o	2025-10-18 19:24:51.919326605 +0530
headers.h	2025-10-18 15:24:36.196068272 +0530
logarithm.c	2025-10-18 19:28:36.927439644 +0530
logarithm.o	2025-10-18 19:28:43.308301924 +0530
Makefile	2025-10-18 18:49:35.055514414 +0530
mul.c	2025-10-17 23:21:22.000000000 +0530
mul.o	2025-10-18 19:24:51.895326607 +0530
power.c	2025-10-18 15:24:23.530424514 +0530
power.o	2025-10-18 19:24:51.953326601 +0530
root.c	2025-10-18 15:25:58.462291304 +0530
root.o	2025-10-18 19:24:51.987326598 +0530
SimpleMath.c	2025-10-18 19:24:35.593412444 +0530
SimpleMath.o	2025-10-18 19:24:51.789326619 +0530
sub.c	2025-10-17 23:21:26.000000000 +0530
sub.o	2025-10-18 19:24:51.871326610 +0530
UserInterface	2025-10-18 19:28:43.346031398 +0530
UserInterface.c	2025-10-18 15:28:50.607994633 +0530
UserInterface.o	2025-10-18 19:24:51.754326622 +0530

```
kaushik@kaushiks-OMEN:~/data/Documents/Programming lab/Make concepts$
```

Figure 5: Modified timestamps after modifying 1 file and doing make

As it can be seen from the above image, “make” compiled only the modified `logarithm.c` file and gave `logarithm.o` and it made a new executable, as it was dependent on `logarithm.o`. Other file’s timestamp remains the same (can be compared with Figure 3 and Figure 5).